

Image:

DSA402_homework2

Load data and run `glimpse()`

```
library(readr)
library(dplyr)

# Column names for the YouTube-BB classification CSV
col_names <- c("youtube_id", "timestamp_ms", "class_id", "class_name", "

# Big - file code:
df <- readr::read_csv(
  "youtube_boundingboxes_classification_validation.csv",
  col_names = col_names,
  n_max = 200000,
  show_col_types = FALSE
)

# Sanity checks
class(df)
```

```
[1] "spec_tbl_df" "tbl_df"      "tbl"        "data.frame"
```

```
dim(df)
```

```
[1] 200000      5
```

```
names(df)
```

```
[1] "youtube_id"      "timestamp_ms"    "class_id"        "class_name"
[5] "object_presence"
```

```
#
dplyr::glimpse(df)
```

Rows: 200,000

Columns: 5

```
$ youtube_id      <chr> "AACebVo-JXY", "AACebVo-JXY", "AACebVo-JXY", "AAC
$ timestamp_ms    <dbl> 283000, 284000, 285000, 286000, 287000, 288000, 2
$ class_id        <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3
$ class_name      <chr> "boat", "boat", "boat", "boat", "boat", "boat", "
$ object_presence <chr> "present", "present", "present", "present", "pres
```

Question 1: What do you notice at first glance when you load the dataset (like the number of rows and columns, presence of missing values, data types)?

- At first glance, the dataset loads as a data frame with 200,000 rows and 5 columns. The variables include a mix of character and numeric types: *youtube_id*, *class_name*, and *object_presence* are character (<chr>), while *timestamp_ms* and *class_id* are numeric (<dbl>). From the missing value check on this sample, all columns have 0 missing values, which suggests the file is clean. The values also look consistent with the expected structure of a YouTube-BB classification dataset: a video ID (*youtube_id*), a timestamp (*timestamp_ms*), and class labeling fields (*class_id*, *class_name*, and *object_presence*).

Question 2: Are there issues that need to be addressed before analysis (like duplicates or unexpected values, data entry errors)?

- The dataset looks clean in the sample. There are 0 exact duplicate rows (*duplicate_rows* = 0), and there are no missing values. The *object_presence* field contains two expected categories: present (**182,043**) and absent (**17,957**), which looks reasonable. One thing to verify before analysis is that the numeric fields fall in expected ranges: *timestamp_ms* ranges from **0 to 3,346,000**, and *class_id* ranges from **0 to 23**, which seems consistent with IDs and timestamps. Overall, no major data-entry errors show up in this sample.

Question 3: Are all variables correctly classified (integer, double, factor, character) or do some need to be converted?

- Some variables should be converted for more appropriate data types. Even though *timestamp_ms* and *class_id* load as doubles, they represent Whole number values, so converting them to integer is more appropriate. Also, *class_name* and *object_presence* are categorical variables, so converting them from character to factor makes sense for summaries. *youtube_id* should remain a character string because it is an identifier.

Question 4: What do the basic summaries reveal about each variable?

- The basic summaries confirm that the dataset is structured as repeated observations tied to video IDs and timestamps. For *timestamp_ms*, the summary shows values ranging from **0 to 3,346,000**, with quartiles around **29,000 (Q1)**, **77,000 (median)**, and **181,000 (Q3)**, and a mean around **144,425**, indicating timestamps are concentrated earlier with a long right tail. For *class_id*, values range from **0 to 23**, with **Q1 = 2**, **median = 7**, **Q3 = 14**, and a mean around **8.857**, suggesting multiple classes appear in the sample. For the categorical variables, *class_name* and *object_presence* are stored as character in the raw import, and *object_presence* includes both present and absent values. Overall, the summaries suggest the data is consistent and usable with minor type conversions.